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Research Article

Deep-Learning-Based Crack Identification and Quantification for Wooden Components in Ancient Chinese Timber Structures

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Cracks exist in the majority of components of ancient Chinese timber structures and have led to serious mechanical property degradation problems, threatening the safety of the whole structures and making the cracks' detection for wooden components a necessity. With the rapid development of intelligent protection technology of cultural buildings, it is important to establish a scientific identification and quantification method for cracks in wooden components to replace traditional manual detection techniques. Deep learning is precisely such an advanced technology. In this study, images of cracked wooden components were first collected from the Yingxian wooden pagoda and the crack characteristics were analyzed. A dataset for crack segmentation was established using a total of 501 images of cracked wooden components, including a training dataset of 450 images and a validation dataset of 51 images. Based on the mathematical principles of deep learning and the fully convolutional neural networks (FCNN), a deep fully convolutional neural network (d-FCNN) model was constructed based on encoding and decoding methodology. Four model indicators, pixel accuracy (PA), average pixel accuracy (mPA), mean intersection over union (mIoU), and F1-score were analyzed to train the model and determine the optimal model parameters, including learning rate, batch size, and epoch. It concluded that the optimal initial learning rate takes the value of 10^{-4}, batch size of 6, and epoch of 100, achieving the average accuracy of 78.8%. Further, based on the pixel's accumulation principle, a quantitative calculation method for crack length and maximum width was proposed. Two cracked wooden columns were prepared, and crack image identification and quantification experiments were conducted to verify the correctness of the constructed d-FCNN model and the proposed crack quantification method. The results show that the model is suitable for crack intelligence detection, identification, and quantification of cracked wooden components.

1. Introduction

Ancient timber structures are precious architectural and cultural heritage in China, having significant historical, cultural, artistic, and scientific value [1]. Protecting ancient timber structures plays a very important role in inheriting excellent traditional Chinese culture. However, after exposure to the environment and loading action for hundreds of years, the vast majority of ancient timber structures in China have experienced different types and shapes of cracks, leading to the degradation of the mechanical properties of the wooden components. Over time, the safety state and mechanical performance of the whole structure will be seriously threatened [2, 3]. Therefore, scientific crack detection, identification, and quantification are crucial for the safety assessment of ancient timber structures with cracked components.

The commonly used crack detection methods for wooden components in ancient timber structures mainly rely on conventional manual measurements, such as the stress wave method, pilo nail technology, impedance meter detection method, and ultrasonic method, et al. [4, 5]. It is not only time-consuming and labor-intensive but also has strong subjectivity in data reading, easily resulting in

significant measurement errors. In addition, working at heights is often dangerous for engineers [6]. With the development of computer science and the widespread application of crack detection models based on image recognition technology [7, 8], the protection of ancient timber structures is moving towards intelligence [9, 10]. Under such background, the establishment of the intelligent crack identification model for cracked wooden components is of great significance for the scientific protection of ancient timber structures. With the development of convolutional neural network (CNN) and image semantic segmentation technology, deep learning-based image processing technology has been made popular in detecting cracks. Intelligent image and sound wave-image joint recognition methods are most commonly used, the details of which can be referred to references [11–14].

In recent several years, the intelligent crack detection of timber structures based on deep-learning image processing technology has also been extensively investigated. Using an improved FCNN, Huang et al. [15] implemented live tree image segmentation. The front end of this technology does not require any processing of the original image, and the back end introduces conditional random fields to refine the image edges. Wang [16] established a classification and recognition model for cracks and knots based on CNN networks to identify and locate the surface damage of ancient wooden components. The results show that the model has more advantages in crack recognition compared to traditional image operator recognition. However, this method belongs to image classification and cannot accurately segment cracks at the pixel level. Ma [17, 18] used the classic object detection YOLO algorithm to identify cracks in ancient timber structures, and 350 generations were trained based on an own-built dataset. The results show that the model can accurately identify the cracks in wooden components, with an accuracy of 91%. Yang [19] analyzed the limitations of nondestructive testing of ancient timber structures based on the characteristics of cracks and proposed that drones equipped with camera instruments and improved adaptive threshold segmentation algorithms should be used to establish a crack monitoring system to achieve long-term monitoring of cracks. Further, Yang [20] established a deformation detection system based on feature point matching algorithms, starting from the overall and local aspects of ancient timber structures, to achieve contactless deformation monitoring of ancient buildings.

Note that the deep learning algorithms have not yet been used to segment and identify cracks at the pixel level and to calculate physical parameters such as crack length and maximum width, et al. in ancient timber structures. In addition, algorithms based on deep learning are mainly focused on the identification of surface cracks in concrete and steel structures, while there is no relevant research on intelligent crack detection for ancient timber structures [21–23]. Also, there is a lack of crack images of ancient wooden components. It is necessary to collect crack images from ancient wooden components and establish corresponding datasets, combined with deep learning training and validation, to reasonably construct a deep learning network suitable for identifying cracks in ancient wooden components.

In this paper, a batch of effective cracked wooden component images were collected through on-site exploration of the Yingxian wooden pagoda, and the characteristics of the cracks and images were summarized. The entire process of dataset production was elaborated in detail on the steps of image collection, cropping, labeling, and format division. Within the framework of deep learning theory, a crack identification model based on the VGG16-Unet network was constructed. Based on the variability of model parameters, the influence of different parameters on model training and validation indicators such as loss value, F1-score, mPA, and mIoU was analyzed. The optimal parameters for crack image recognition were determined, and the recognition effect was compared with other segmentation methods. Based on the binary crack images, a theoretical calculation formula for determining the crack length and maximum width of cracked wooden components was proposed.

2. Characteristics and Dataset of Cracks in Wooden Components

Cracks in different ancient wooden components generally have many common characteristics. To automatically identify and detect cracks by computers, it is necessary to have computer programs learn these characteristics and classify cracks with different numbers and shapes to lay the foundation for crack identification and quantification. Thus, many images of cracked wooden components were collected from the Yingxian wooden pagoda in Shuozhou, Shanxi Province, and the characteristics of cracks were systematically analyzed to create dataset.

2.1. Image Characteristic of Cracks in Wooden Components. The images of cracked wooden components were manually captured using industrial cameras, brand Sony-6300, with a resolution of 6000 pixels × 4000 pixels and a focal length of 50 mm. Due to the enclosed environment and insufficient lighting at the construction site, LED spotlights were used to supplement the light source. To ensure the diversity of cracks, different cracked components such as wooden beams, columns, and Dou-Gongs were selected to shoot under different lighting conditions and angles. For cracked wooden components at higher positions, the wooden staircase inside the Yingxian Wooden Pagoda, steel platforms, ladders, and selfie poles were used to assist in shooting.

Due to the fiber-like composition of wood, the crack morphology of ancient wooden components is different from that of concrete and steel components and can be summarized as follows: (1) usually narrow and long, with a large aspect ratio; (2) mostly being slender at both ends and widest in the middle; (3) a few cracks have larger width at one end and gradually extend to the smaller end; (4) usually multiple cracks coexist, appearing as a continuous whole and locally connected in a filamentous manner.

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Efficient and accurate apparent cracks' identification in wooden components requires accurate extraction of crack features, training of neural network models, continuous parameter optimization, and adjustment to obtain binary images of cracks after semantic segmentation, so as to further measure their physical values. The image characteristics of the cracked wooden components are summarized as follows: (1) the image light is uneven, overexposed, or weak; (2) the contrast between image crack areas and noncracked areas is obvious and easy to distinguish; (3) there are intermittent connections or fractures between image cracks; (4) the image cracks are slender and the width of the cracks is much smaller than the length; (5) the shooting angles vary from the front and elevation angles; (6) there are external environmental defects such as dust and human damage on the surface of the component, as shown in Figure 1. Note that the above typical characteristics of cracks should be considered when collecting cracked wooden components and establishing the dataset of crack images.

2.2. Crack Dataset of Wooden Components. Crack dataset is the foundation for establishing a deep-learning-based model for identifying apparent cracks in ancient wooden components. The quality of crack images directly affects the training and validation accuracy of the deep learning model. Therefore, the crack image selection should follow the principles of the nonblurriness image, clear and visible cracks, and cracks as much as possible in the middle of the image. After screening the collected crack images of Yingxian wooden pagoda components, 501 images were obtained as the original images to create the crack dataset of the wooden component. The crack images were stored in JPG format, providing samples for the crack dataset establishment.

2.2.1. Image Cutting. The varying sizes of the captured images usually have a negative impact on the training speed and accuracy of the neural network. Additionally, due to shooting conditions, some crack images are not located in the middle of the photo. Therefore, it is necessary to trim the original image to preserve the crack area, remove the noncracked interference information, and number the cropped images from 0 to 500. The images were cropped using PicPick software into square images' pixel size of 256 ×256. The cropping process is shown in Figure 2.

2.2.2. Image Annotation. The open-source image annotation tool *LabelMe* software was used to segment and annotate the crack images. During labeling, multiple cracks in the same wooden component image can be labeled simultaneously. The details of image annotation directly affect the training accuracy of the network model. Thus, during image annotation, the points on the crack boundary should be uniformly selected and the boundary should be as close to the original curve trajectory as possible without omission. The contour of the crack through irregular rectangles connected at the beginning and end was drawn, and a file in *.json* format was generated to save the crack information. The generated *.json* files were converted into the mask data required for model training through the *json_to_dataset* function. Each *.json* file generates a folder, where *img.png* and *Label.png* are the original crack images of wooden components and the label mask file, respectively, as shown in Figure 3.

The dataset, abbreviated as ATM-Crack-Data, used 10-fold cross-validation and was divided into two parts: the training set and the validation set, with a ratio of 90% and 10%, respectively. The training set includes 450 photos, while the validation set has 51 photos.

3. VGG16-Unet-Based Network Model for Crack Image Segmentation

3.1. Model Establishment. Based on the CNN theory, a fully connected layer was removed and a deconvolution layer was added. An FCNN was used for crack recognition and segmentation of ancient wooden component images. The main difference between CNN and FCNN is that the latter replaces the fully connected layer of the former with a convolutional layer while adding a deconvolution layer to achieve end-to-end training of the model, where both input and output are images.

Due to the obvious characteristics and small sample size of crack images of ancient wooden components, a neural network model suitable for identifying crack images in ancient wooden components was established based on the VGG16-Unet network, hereinafter referred to as the ATM-Crack-U-net model. The crack segmentation network model of wooden components included a shrinkage path and a symmetric expansion path. The crack image continuously extracts crack features through the shrinkage path and enlarges feature points in the symmetric expansion path to obtain an image that only contains crack features. The structure of the ATM-Crack-U-net model is illustrated in Figure 4.

It is worth noting that the VGG16 neural network model was proposed by Simonyan and Zisserman from the University of Oxford in 2014 [24]. The model consists of 16 layers, and the convolutional layer uses 3×3 convolutional kernels, step size 1, pooling layer 2×2. The model structure after replacing the convolutional layer is illustrated in Figure 5.

The VGG16-Unet neural network uses the first 13 convolutional layers and the first 4 pooling layers of the VGG16 model (i.e., removing 1 pooling layer, 3 fully connected layers, and 1 output classification layer). Convolutional and pooling layers are used during the model contraction path, which is downsampling, while deconvolution and feature aggregation layers are used during the model expansion path, which is upsampling.

After the model construction was completed, the training and validation sets of the data were introduced to the model for training and validation to obtain a crack recognition model, and the validation set was introduced for model testing. The specific process is shown in Figure 6.

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(a) (b) (c)

(d) (e) (f)

Figure 1: Image characteristics of cracks in wooden components. (a) Uneven light. (b) Obvious crack. (c) Intermittent fracture. (d) Slim cracks. (e) Elevation shooting. (f) Environmental interference.



(a) (b)

Figure 2: Image cutting schematic. (a) Before cutting. (b) After cutting.

3.2. Model Parameters and Environment Configuration

3.2.1. Model Parameter Settings. In the training of F-CNN, three hyperparameters, learning rate, batch size, and iteration number were selected as variables to determine the optimal values. Four learning rates were adopted, namely, 10^{-3}, 10^{-4}, 10^{-5}, and 10^{-6}. During the training process, a ladder attenuation strategy was adopted for the initial learning rate to ensure that the model can converge timely. Four batch sizes were adopted, namely, 2, 4, 6, and 8, to improve the accuracy of the model calculation and make the training process smoother without significant oscillations. Two types of iterative training times, 50 and 100 were selected, respectively. In the first half of the network training, the backbone extraction network was frozen, which was the contraction path. The use of pretraining weights will damage the weight file in the early training stage, which can accelerate the training speed. In the second half of the network

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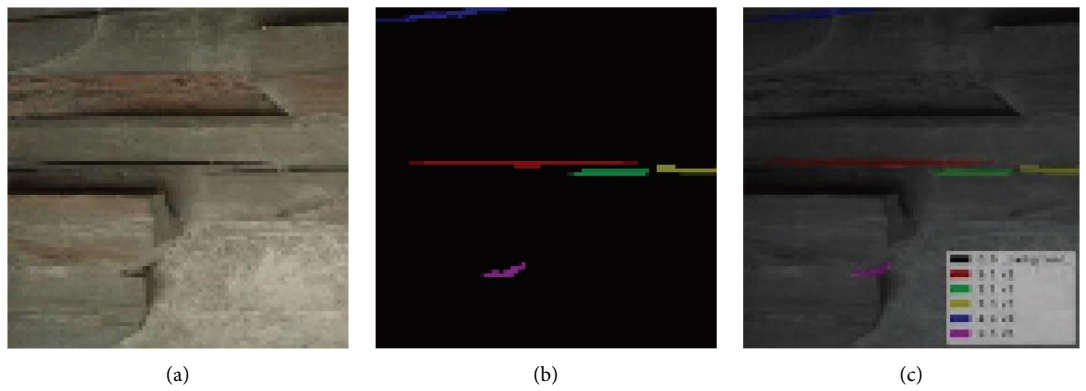


FIGURE 3: Annotated image after *.JOSN* file conversion. (a) RGB. (b) Label. (c) Label_viz.

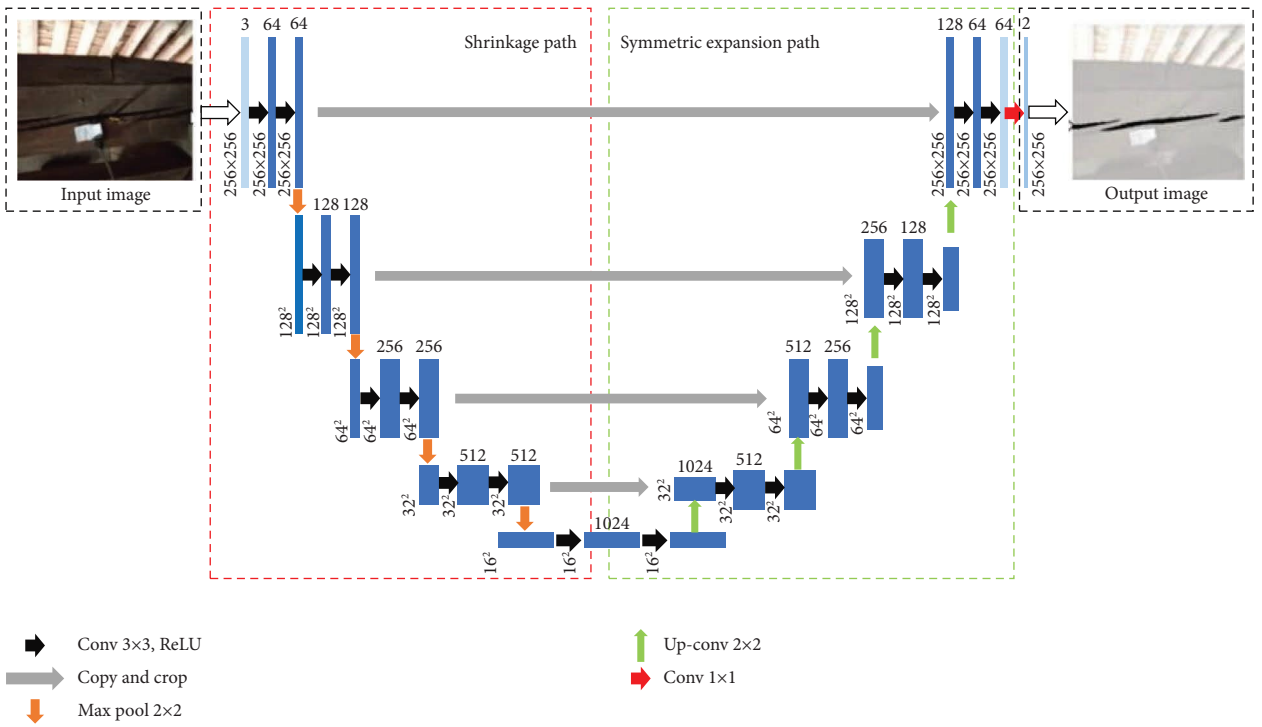


FIGURE 4: Structure diagram of the ATM-Crack-U-net network model.

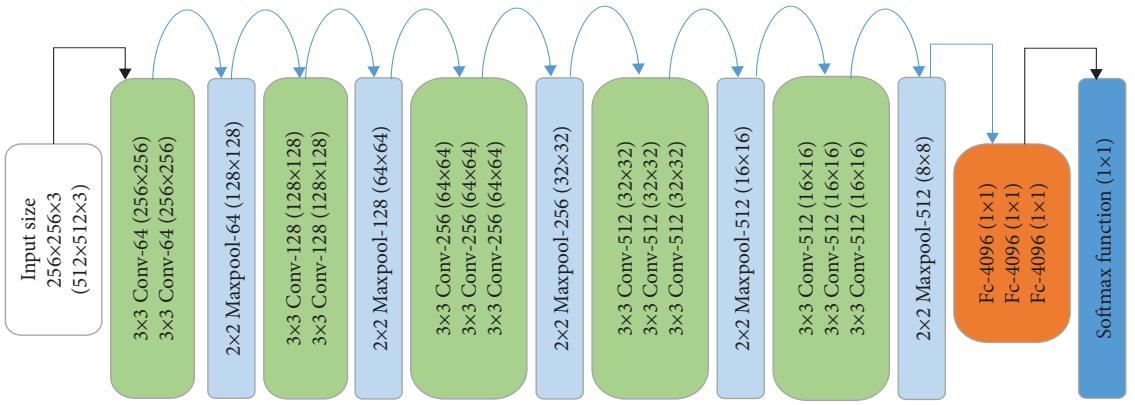


FIGURE 5: VGG16 convolutional neural network structure diagram.

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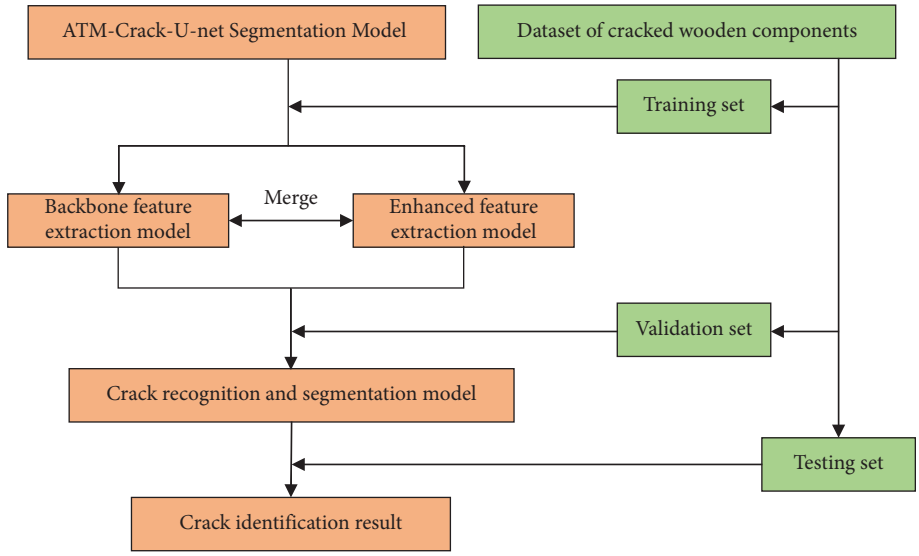


Figure 6: Identification process for cracks in ancient wooden components.

training, thaw the backbone extraction network parameters and perform crack feature superposition training on the symmetric expansion path to extract the combined crack features in more detail. The crack segmentation model for ancient wooden components is a binary classification problem of crack background, so a binary cross entropy loss function was adopted. The Adam adaptive optimization algorithm, which is widely used in neural network training, was used. The ReLU activation function was used to give nonlinear characteristics to the neural network training process, making the function converge faster. A parameter initialization method based on variance scaling was selected. He-initialization method was selected to prevent extreme situations such as vanishing and exploding, inconsistent input and output variances, and others.

3.2.2. Environmental Configuration. The model code was written using Python language, and the experimental virtual environment was built on a Windows system for training, validation, and testing of crack recognition models for ancient wooden components. The configuration steps for the software environment were as follows: (1) Anaconda software was installed and Python 3.6 experimental environment was created, which can visually manage the libraries used in Python language; (2) installing the GPU version of TensorFlow allows for accelerated training of the model using a graphics processor or graphics card; (3) CUDA and cuDNN libraries were installed.

3.3. Model Indicators. After predicting and segmenting crack images using an FCNN model, it is necessary to evaluate the quality of the model. In addition to visual evaluation, quantitative objective model indicators such as PA, mPA, mIoU, and F1-score can be used.

Pixel accuracy, PA, is defined by the proportion of correctly predicted crack pixels to the total pixels, as shown in equation (1).

$$PA = \frac{\sum_{i=0}^{k} p_{ii}}{\sum_{i=0}^{k}\sum_{j=0}^{k} p_{ij}}. \tag{1}$$

Mean pixel accuracy, mPA, is defined as the accumulated and average predicted correct pixel proportions for each category, as shown in equation (2).

$$mPA = \frac{1}{k+1}\sum_{i=0}^{k}\frac{p_{ij}}{\sum_{j=0}^{k} p_{ij}}. \tag{2}$$

Mean intersection over union, mIoU, refers to the ratio of the intersection area of two different regions to their union area. These two regions are the true value region and the predicted value region, respectively. After calculating the intersection over union on each category, average it, as shown in equation (3).

$$mIoU = \frac{1}{k+1}\sum_{i=0}^{k}\frac{p_{ij}}{\sum_{j=0}^{k} p_{ij} + \sum_{j=0}^{k} p_{ji} - p_{ii}}, \tag{3}$$

where p_{ij} represents that the ith class of pixels is incorrectly predicted as a class j; p_{ii} represents that the ith class of pixels is incorrectly predicted as a class i; k represents the total number of categories.

F1-score is defined in the form of weighted accuracy and recall, as shown in equation (4).

$$F1 = \frac{2 \times P \times R}{P + R}, \tag{4}$$

where the accuracy P is defined as the proportion of pixels that the model considers to be cracks and actually cracks to pixels that the model considers to be cracks, used to measure the accuracy of predicting cracks; the recall rate R is defined as the proportion of pixels that the model considers to be cracks and are actually cracks to pixels that are actually cracks, used to measure the completeness and coverage of crack prediction. If it is too low for P, the identification of

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cracks in the wooden components is not accurate enough, and if it is too low for R, many crack areas in the wooden components are not distinguished. Accuracy and recall rate are defined as

$$P = \frac{TP}{TP + FP}, \tag{5}$$

$$R = \frac{TP}{TP + FN}, \tag{6}$$

where TP represents the correctly classified positive sample, where the model distinguishes pixels into crack points and is correctly extracted as pixels in the crack area; TN is the correctly classified negative sample, where the model distinguishes pixels as background points and is correctly extracted as pixels in the background area; FP is a negative sample that has been misclassified, where the model identifies background pixels as crack points and mistakenly extracts them as background area pixels; FN is a positive sample that has been misclassified, where the model identifies crack pixels as background points and mistakenly extracts them as pixels in the crack area.

3.4. Model Training

3.4.1. Learning Rate Attenuation Comparison under Different Batch Sizes. When training the crack identification model for ancient wooden components, the backbone extraction network was used for 50 generations of freezing training and 50 generations of thawing training, with a total of 100 iterations. The model was trained using four initial learning rates of 10^{-3}, 10^{-4}, 10^{-5}, and 10^{-6}, and four batch sizes 2, 4, 6, and 8.

During the training process, the ladder attenuation method was applied for the learning rate attenuation strategy, meaning that the learning rate continuously decayed from the initial value according to the transformation of the training situation and the current learning rate was half of the previous learning rate. In the programming, it was specified that when the validation set loss does not decrease after two iterations of training, the learning rate will decay to half of the original value before continuing with training. The learning rate decay curves for the four learning rates and four batch sizes at 100 iterations are shown in Figure 7.

It can be seen from Figure 7 that the learning rate decay of the crack identification model for ancient wooden components is as follows: whether the model is frozen or thawed, as the initial learning rate decreases, the number of decay times decreases. It indicates that the number of decay times increases with a higher learning rate, which can accelerate the learning rate and find the global optimal value as soon as possible. On the contrary, when the learning rate is small, it is easy to fall into local optimal values. As the initial learning rate decreases, the first decay Epoch of the learning rate continuously increases, indicating that a larger learning rate can accelerate the model convergence process in the early training stage.

3.4.2. Loss Values Comparison under Different Batch Sizes. To analyze the impact of batch sizes on the loss value reduction of the crack identification model, 100 iterations were set with batch sizes of 2, 4, 6, and 8. The loss value curves of the crack identification model with different batch sizes in the training set and validation set are compared and shown in Figure 8 and Supplemental Figures 1–3.

When the initial learning rate is 10^{-3}, the loss curves are shown in Figure 8. During the training and validation of the model backbone network at the 50th epoch, the loss values rapidly increased from 0.2 to 0.8, continued to fluctuate and tended to remain unchanged, indicating that the initial learning rate was improperly set. After iterating through the 20th epoch, the loss of the validation set was larger than that of the training set, indicating that the model had preliminary recognition ability. Compared with Supplemental figure 1, it can be seen that the loss values of the model briefly increased to 0.3 at the 50th epoch iteration and gradually decreased, indicating that the initial learning rate of 10^{-4} was set more reasonably than 10^{-3}. Compared to Supplemental figure 2, it can be seen that the loss values decreased smoothly before the 50th epoch, meaning that the initial learning rate of 10^{-3} was too high and needed to be reduced.

When the initial learning rate is 10^{-4}, the loss curves are shown in Supplemental figure 1. During the freezing training process of the backbone network model, after the 10th epoch, the training loss values all decreased to below 0.2, verifying that the loss oscillated around 0.2. When the model was thawed from the 50th epoch to the 100th epoch, and the batch size was 2, the loss values remained fluctuating at around 0.2 after the 70th epoch. When the batch size was 4, the loss values remained fluctuating around 0.18 after the 75th epoch. When the batch sizes were 6 and 8, the loss oscillated around 0.16 after the 80th epoch. Therefore, when the initial learning rate was 10^{-4}, as the batch size increased, the training and validation loss values continued to decrease. After increasing the batch size to 6, the loss of the model did not decrease significantly.

When the initial learning rate is 10^{-5}, the loss curves are shown in Supplemental figure 2. During the freezing training process of the backbone network, the training and validation loss values were both above 0.2. During the thawing training process, as the batch sizes increased, the loss values of the validation set fluctuated around 0.2 with continuous iteration in the later stage, without a significant downward trend. Compared to Supplemental figure 1, from the perspective of loss values, the training effect at an initial learning rate of 10^{-5} was not as good as at 10^{-4}.

When the initial learning rate is 10^{-6} and the batch size increased from 2 to 8, the loss curves are shown in Supplemental figure 3. During the training and validation process of the model, the loss values were above 0.2 and there was large difference between the predicted value and the true value obtained by the crack model, indicating that the initial learning rate 10^{-6} was relatively low and needed to be increased.

The comparison curves of the optimal loss values are shown in Supplemental figure 4. It can be seen that an initial learning rate of 10^{-6} make the worst in training and validation results, indicating that this value was unreasonably

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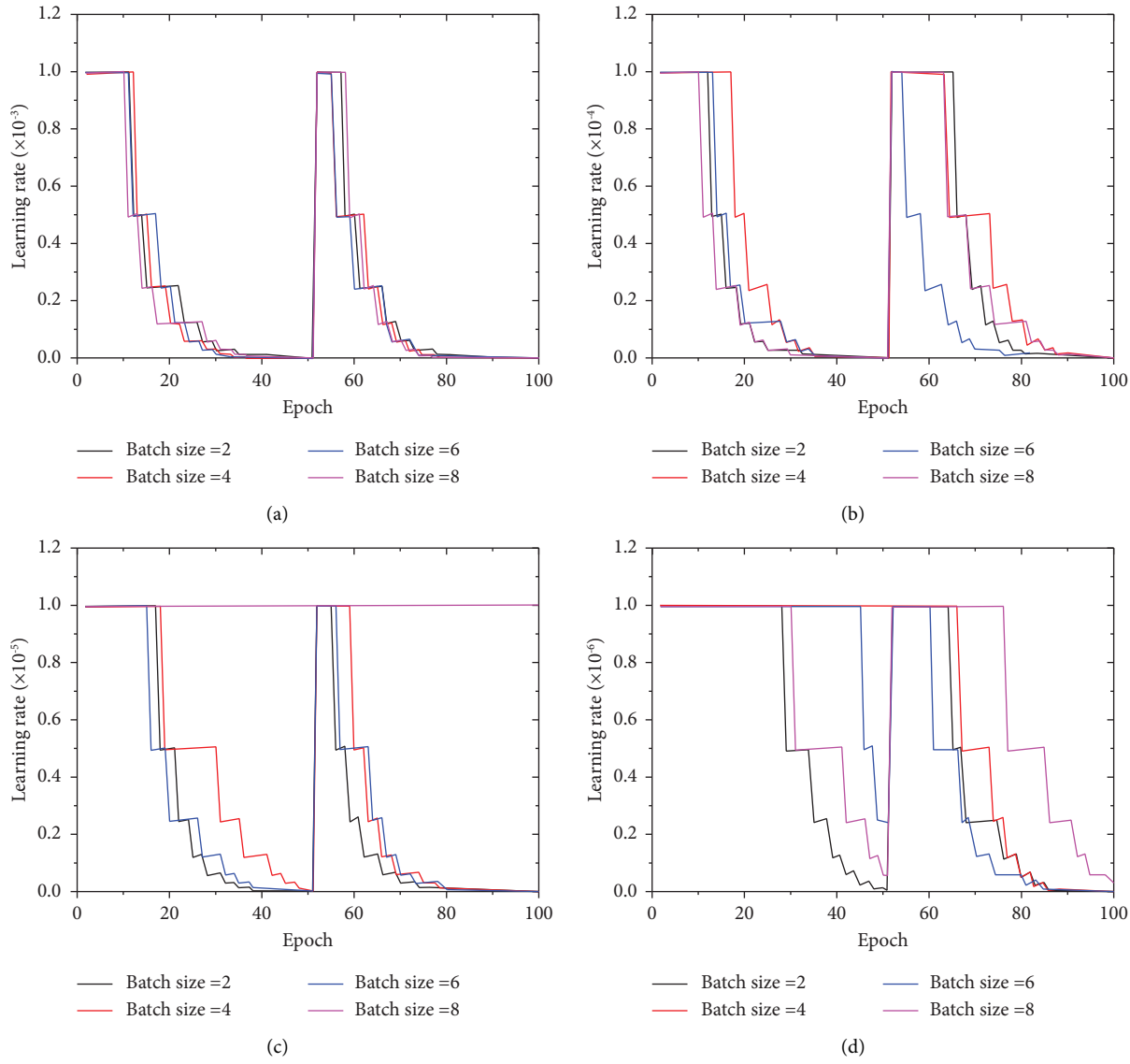


FIGURE 7: Decay of learning rate for each batch after 100 iterations. (a) Learning rate 10^{-3}. (b) Learning rate 10^{-4}. (c) Learning rate 10^{-5}. (d) Learning rate 10^{-6}.

set. The optimal loss values at an initial learning rate of 10^{-4} on both the training and validation sets were smaller than those at 10^{-5}, meaning that 10^{-4} was the optimal learning rate. In addition, seen from the perspective of effective initial learning rates of 10^{-4} and 10^{-5}, when the batch size was 6, the optimal loss value for model training and validation was the smallest, indicating that 6 was the optimal batch size.

3.4.3. F1 Values' Comparison for Different Batch Sizes. To analyze the influence of different initial learning rates and batch sizes on the F1-scores of the crack identification model for ancient wooden components, 100 iterations were set, with initial learning rates of 10^{-3}, 10^{-4}, 10^{-5}, and 10^{-6}, and batch sizes of 2, 4, 6, and 8. The F1-score curves for model training and validation are shown in Figure 9 and Supplemental figures 5–7.

When the initial learning rate was 10^{-3}, the F1-scores (Figure 9) of the model backbone network rapidly decreased from 0.83 to around 0.5 during thawing training and validation at the 50th epoch and continuously oscillated towards 0.5, indicating that the initial learning rate 10^{-3} was inappropriate. After the 20th epoch, the F1-scores of the validation set was smaller than the training set, indicating that after training for 20 generations, the model had preliminary recognition ability. Compared to Supplemental figure 5, the F1-scores of the model briefly decreased to 0.75 after the 50th epoch and gradually increased, indicating that the initial learning rate of 10^{-4} was reasonable compared to 10^{-3}. Compared to Supplemental figure 6, the F1-scores increased normally after 50th epoch. Therefore, the initial learning rate of 10^{-3} was relatively high and needed to be reduced.

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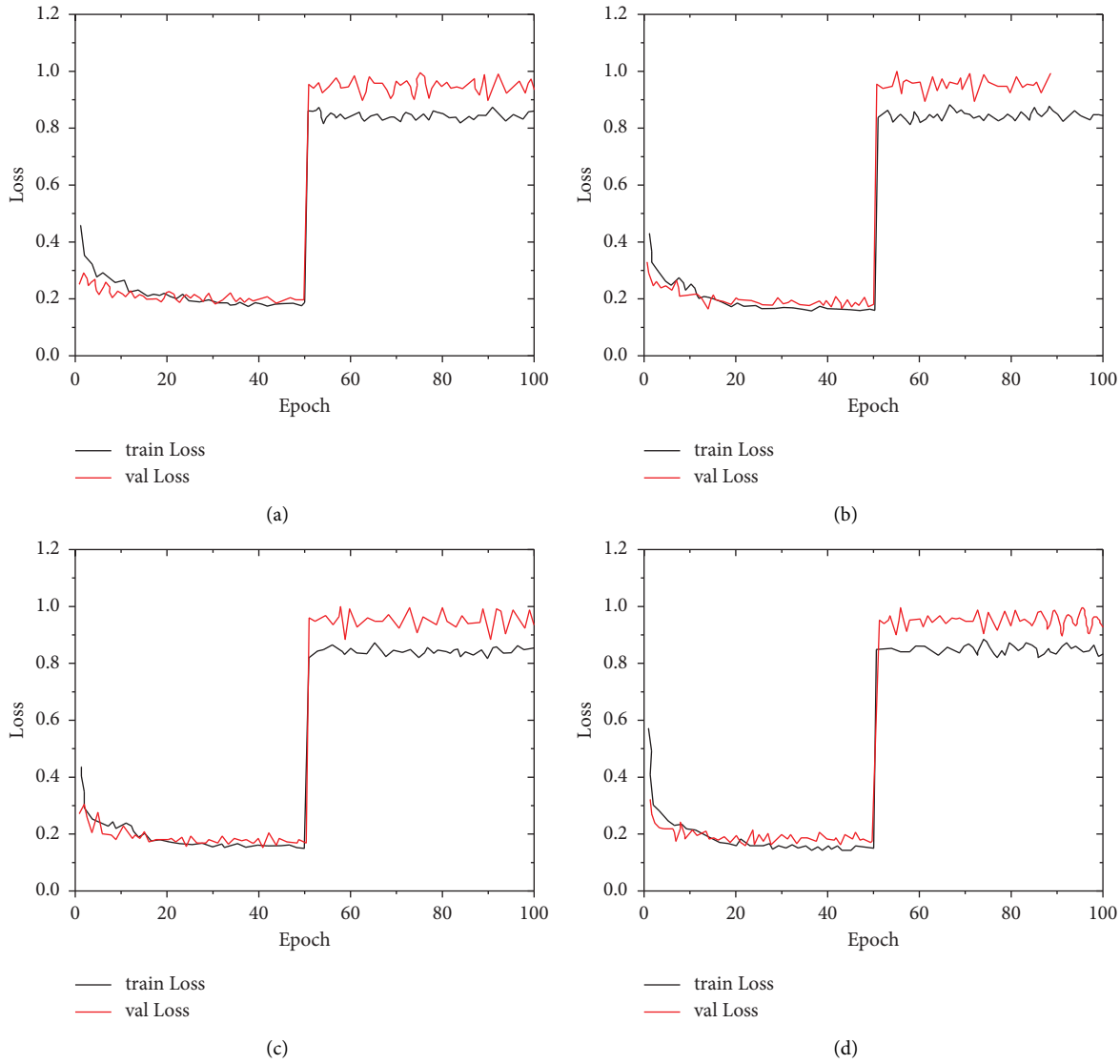


FIGURE 8: Loss values under different batches when $Lr = 10^{-3}$. (a) Batch size = 2. (b) Batch size = 4. (c) Batch size = 6. (d) Batch size = 8.

When the initial learning rate was 10^{-4}, see Supplemental figure 5, during the freezing training process of the backbone network, the F1-scores raised to above 0.8 and ranged from 0.8 to 0.9 after 5th epochs of training. When batch size was 2, the F1-scores remained fluctuating at around 0.82 after the 65th epoch. When the batch size becomes 4, the F1-scores oscillated around 0.84 after the 70th epoch. The F1-scores oscillated around 0.86 after the 75th epoch when the batch sizes were 6 and 8. It can be seen that when the initial learning rate was 10^{-4}, the F1-scores of training and validation continued to decrease as the batch size increased in the later stage. When the batch size increased to 6 and larger, the F1-score did not increase significantly.

When the initial learning rate was 10^{-5}, see Supplemental figure 6, during the freezing training process of the backbone network, the F1-scores for training and validation sets fluctuated around 0.8. As the batch processing increased and the validation set continued to iterate in the later stages, the F1-scores fluctuated between 0.8 and 0.85. Analyzing the F1-scores in Supplemental figure 5, it was found that the F1-scores at the initial learning rate of 10^{-5} were not higher than that at 10^{-4}.

When the learning rate was 10^{-6}, see Supplemental figure 7, as the number of iterations increased, the F1-scores continuously increased during the early iterations and oscillated around 0.8 during the later iterations. Comparing Supplemental figures 5-6, it can be seen that the F1-scores with initial learning rates of 10^{-4} and 10^{-5} were close to 0.9, both of which were higher than those at 10^{-6}, indicating that the initial learning rate at 10^{-6} was set unreasonable.

Under different operating conditions, there was always a set of optimal weight files during the training and validation process of the ATM Crack U-net network model. The comparison curves of the optimal F1-score for different conditions are shown in Supplemental figure 8. It can be seen that when the initial learning rate was 10^{-6}, the model F1-scores were all the lowest compared with that under other

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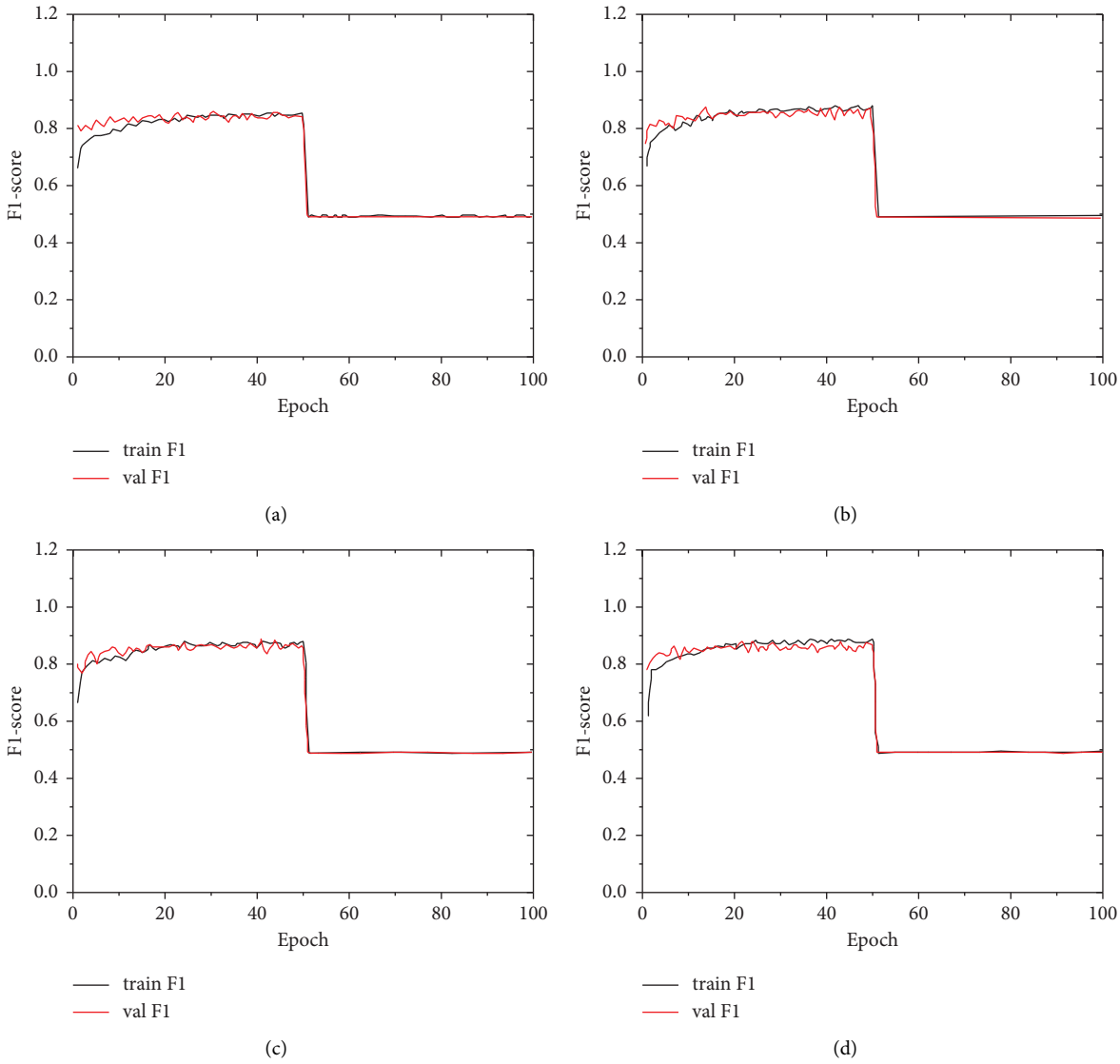


FIGURE 9: Predicted difference between different batches when $Lr = 10^{-3}$. (a) Batch size = 2. (b) Batch size = 4. (c) Batch size = 6. (d) Batch size = 8.

initial learning rates, indicating that the initial learning rate of 10^{-6} was unreasonable. When the initial learning rate was 10^{-3}, the F1-scores were fully thawed after the 50th epoch and the F1-values dropped sharply to 0.5, indicating that the initial learning rate 10^{-3} was set too high and unreasonable.

In addition, when the batch size was the same, the optimal F1-scores at an initial learning rate 10^{-4} on both the training and validation sets were greater than that at 10^{-5}, implying that 10^{-4} was the optimal learning rate. From the effective initial learning rates of 10^{-4} and 10^{-5}, it can be seen that when the initial learning rates were the same, the optimal F1-scores for training and validation were the highest when the batch size was 6. Thus, 6 is the optimal batch size.

In summary, the superiority of the upward trend analysis model based on F1-scores indicated that the higher the F1-score, the better the balance between the accuracy and recall of the model at a higher level, and the better the performance of the model. Therefore, selecting the optimal initial learning rate can effectively improve the excellence of crack recognition in ancient wooden components. With the optimal learning rate, selecting an appropriate batch size can significantly improve the F1-score of the model.

3.4.4. Model Indicators Comparison under Different Iterations. From the above analysis, it can be seen that excluding the initial learning rate of 10^{-3}, the model was set with 50 iterations, with initial learning rates of 10^{-4}, 10^{-5}, and 10^{-6}, batch sizes of 2, 4, 6, and 8, and a total of 12 parameter combinations for model training and validation. The relevant indicators are shown in Table 1.

From Table 1, it can be seen that when the number of iterations of the model increased from 50 to 100, the loss of the validation set only increased in 2 working conditions, while the rest decreased to different degrees. Among them, when the initial learning rate was 10^{-4} and the batch size was

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Table 1: Comparison of indicators with different iterations.

Batch size	Learning rate	Val loss			Val F1-score		
		Epoch = 50	Epoch = 100	Percentage reduce (%)	Epoch = 50	Epoch = 100	Percentage increase (%)
2	10^{-4}	0.159	0.170	−1.1	0.879	0.873	0.6
	10^{-5}	0.175	0.185	−1.0	0.866	0.859	0.7
	10^{-6}	0.233	0.224	0.9	0.828	0.830	0.2
4	10^{-4}	0.165	0.148	1.7	0.879	0.888	0.9
	10^{-5}	0.184	0.172	1.2	0.864	0.873	0.9
	10^{-6}	0.232	0.216	1.6	0.835	0.845	1.0
6	10^{-4}	0.157	0.139	1.8	0.886	0.896	1.0
	10^{-5}	0.170	0.167	0.3	0.880	0.878	−0.2
	10^{-6}	0.260	0.245	1.5	0.811	0.827	1.6
8	10^{-4}	0.161	0.148	1.3	0.879	0.883	0.4
	10^{-5}	0.184	0.173	1.1	0.864	0.871	0.7
	10^{-6}	0.229	0.227	0.2	0.830	0.833	0.3

6, the maximum decrease was 1.8%. For the F1-scores in the validation set, only one working condition decreased, while the rest increased to varying degrees. When the initial learning rate was 10^{-6} and the batch processing was 6, the maximum increase was 1.6%. In summary, a moderate increase in the number of iterations of the model can effectively reduce the loss value and increase the F1-score, thereby improving the excellence of the model.

3.5. Model Indicator Validation

3.5.1. mIoU and mPA Analysis for Different Parameters. To study the recognition accuracy of crack segmentation models for ancient wooden components under different initial learning rates, batch sizes, and iteration times, two iteration parameters were set as 50 epoch and 100 epoch, totaling 12 working conditions. After the training of each working condition, 50 or 100 weight files were obtained. The weight file with the lowest loss in the validation set was selected as the optimal weight and was loaded into the model for the current working condition. The mIoU and mPA were calculated to evaluate the recognition performance of the model, see Table 2.

When the batch size was the same, as the initial learning rate decreased, the model indicators mIoU and mPA decreased, having a positive correlation. When the initial learning rate was the same, there was no significant correlation between batch size and model indicators mIoU and mPA.

When the initial learning rate was 10^{-4} and the number of iterations changed from 50th epoch to 100th epoch, the recognition indicators mIoU and mPA were improved. For example, when the batch size was 8, mIoU increased by 2.07 and mPA by 1.8, making the model more accurate in identifying cracks in ancient wooden components.

When the parameter combination was at an initial learning rate of 10^{-4}, batch processing capacity of 6, and iteration of 100 epochs, the mIoU and mPA of the model were 69.59 and 78.83, which were the best recognition indicators for the model.

3.5.2. mIoU and mPA Analysis for the Optimal Parameters. During the iteration process of 100 generations, 100 models were validated one by one on the validation set of the Yingxian Wooden Pagoda dataset. The mIoU and mPA transformations of the model were evaluated from the initial training stage to the end stage, and the best indicator model under this parameter was found.

The larger the values of mIoU and mAP indicators, the better the model recognition effect. The model weight files were numbered in the form of ep001-loss0.430-val_loss0.316 (with a training set loss value of 0.430 and a testing set loss of 0.316 for the first iteration generation), and a total of 100 weight files were obtained after training. The validation set of the Yingxian Wooden Pagoda dataset consisted of 51 crack images. After comparing the crack segmentation of each validation set image with the LabelMe annotated mask image, the crack, background, and overall segmentation indicators were obtained, as shown in Figures 10 and 11.

From Figures 10 and 11, it can be seen that in the early stages of freezing and thawing training, the crack identification index was unstable, with the highest mIoU of the crack reaching 68.28 and the lowest dropping to 47.86. The mPA reached a maximum of 85.62 and dropped to a minimum of 69.37. When the model froze and trained in the later stages of the 25th to 50th generation, the mean mIoU and mPA identification indicators for model cracks were 67.5 and 76.2, respectively. In the 75th to 100th epoch of post-thawing training, the mean value of the model crack recognition index mIoU was 68.1, an increase of 0.6 compared to freezing training, and the mean value of mPA was 77.8, an increase of 1.6 compared to freezing training. In these two stages, the mIoU and mPA of the model reached a flat value, at which point the model identification was relatively stable and the accuracy of crack identification was high.

3.6. Model Testing of Crack Images. In this section, the established ATM-Crack U-net model was tested through a testing dataset, covering the characteristics of cracks in ancient wooden components, to ensure good applicability and robustness of the model to identify different types of cracks.

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Table 2: Verification set crack segmentation index.

Batch size	Learning rate	mIoU			mPA		
		Epoch = 50	Epoch = 100	Percentage reduction (%)	Epoch = 50	Epoch = 100	Percentage increase (%)
2	10^{-4}	69.33	69.55	0.22	77.20	77.30	0.1
	10^{-5}	66.82	67.06	0.24	74.75	76.09	1.34
	10^{-6}	59.35	59.17	−0.18	75.07	72.04	−3.03
4	10^{-4}	68.05	69.15	1.1	77.06	77.49	0.43
	10^{-5}	65.76	65.54	−0.22	72.85	74.02	1.17
	10^{-6}	59.76	58.09	−1.67	74.64	73.15	−1.49
5	10^{-4}	68.32	69.59	1.27	75.98	78.83	2.85
	10^{-5}	65.34	65.9	0.56	74.19	73.47	−0.72
	10^{-6}	53.79	51.05	−2.74	70.77	65.48	−5.29
6	10^{-4}	66.99	69.06	2.07	76.37	78.17	1.8
	10^{-5}	65.98	65.09	−0.89	77.32	72.98	−4.34
	10^{-6}	57.98	57.14	−0.84	70.96	72.25	1.29

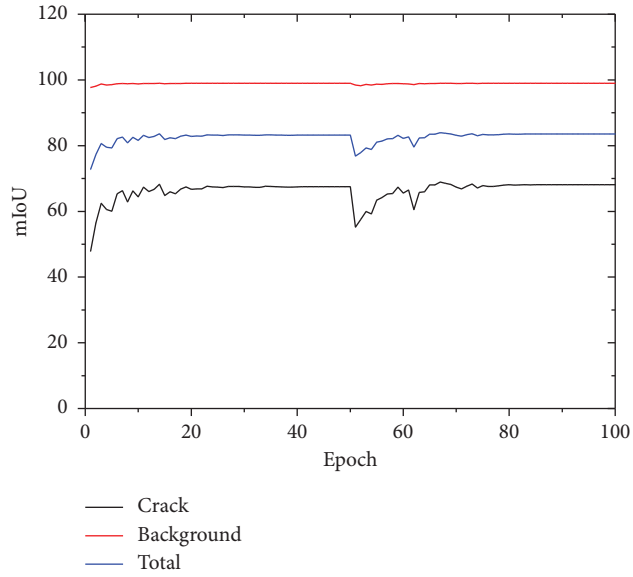


Figure 10: mIoU of each model under optimal parameters.

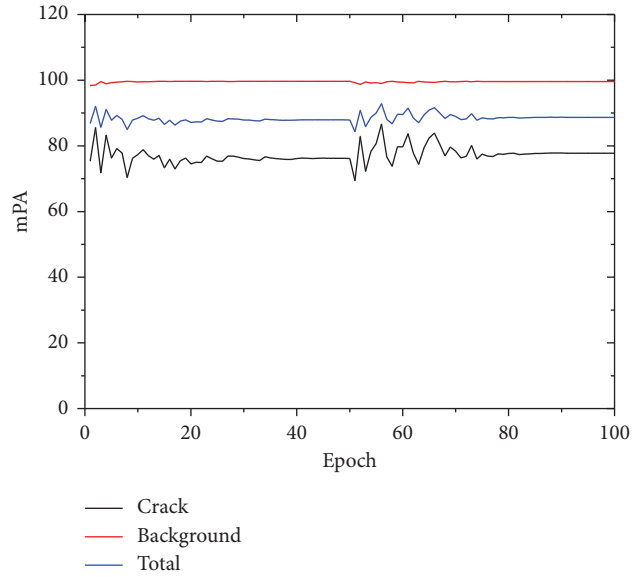


Figure 11: mPA of each model under optimal parameters.

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Five pictures of cracks in wooden components were selected and numbered P1-P5 in sequence. Among them, three pictures were taken from the Yingxian wooden pagoda: P1, having a transverse crack with a reduced width from left to right, suitable background light, and presence of interference objects such as selfie rods; P2, showing an oblique crack with a thin and long middle width at both ends, a large middle width at both ends, a dark background light, and interference objects such as selfie rods; P3, having a transverse crack with a thin and long middle width at both ends, good background light, and the presence of interference objects such as steel tape. A photo of a cracked wooden beam in a certain residential building, P4, shows a transverse through crack, with an overall slender and uniform width, suitable background light, and the presence of white wire tubes, electric wire bulbs, and other interfering substances. A photo of a laboratory wooden column specimen, P5, shows a vertical through crack, with an overall slender and uniform width, good background light, and the presence of wooden knots around the crack and interfering substances such as strip iron plates in the background.

The optimal parameters, initial learning rate of 10^{-4}, batch size of 6, and iteration of 100 generations for the model, were used. The weight file ep095-loss0.167-val_-loss0.169 was selected as the tested model under 95th epoch to test the five cracked wooden component images. The test results are shown in Figure 12. It can be seen that the established model overall had a good recognition effect on the cracks in wooden component, can accurately segment the crack contour, provide binary images of cracks, and achieve intelligent segmentation of cracks from wooden component images.

To further illustrate the effectiveness of the crack segmentation method established in this study for wooden components, watershed segmentation algorithm (WSA), adaptive iterative threshold method (OSTU), Canny operator, and ATM-Crack U-net model were used to test the same batch of testing images P1-P5. The specific test results are shown in Figure 13.

It can be noted that the WSA and OTSU algorithms had poor recognition performance for cracks in wooden components. During the process of crack recognition, it was severely affected by the illumination of the image and cracks can hardly be identified for wooden component images with poor and dark lighting. For objects with backgrounds such as selfie poles, white wire tubes, and similar cracks, the contour of the cracks will be displayed during the recognition process, causing significant errors in recognition.

In the process of identifying cracks using the Canny method, the threshold was continuously adjusted to achieve the most significant expression of the crack contour. From the recognition effect of P1-P5, traditional image recognition was not effective in identifying cracks in wooden components, with many noise points causing blurry crack contours. Thus, this method was not suitable for wooden component crack recognition and segmentation.

The ATM-Crack-U-net model constructed in this study had a very good effect on identifying cracks in wooden components. The crack segmentation results were noise-free, the crack contour was smooth, the light intensity had no impact on the recognition process, and the crack can also be segmented in complex backgrounds.

In summary, the crack recognition and segmentation algorithm proposed in this study applied to the identification of cracks in ancient wooden components and had more advantages compared to conventional image recognition algorithms.

4. Quantitative Method for Cracks in Wooden Components

4.1. Sample Shooting. The experiment was completed at the Key Laboratory of Structural Engineering and Earthquake Resistance of the Ministry of Education at the Xi'an University of Architecture and Technology. Two wooden column specimens, numbered "01" and "02," with a total length of 800 mm and a diameter of 100 mm were selected. "01–300" represented the image of the "01" specimen taken at a distance of 300 mm. The "01" specimen had two cracks named A and B, with crack lengths of 510 mm and 400 mm, respectively; the "02" specimen had only one crack named C, with a crack length of 605 mm. The specimen is shown in Figure 14. Six sets of object distances of 0.3 m, 0.5 m, 0.8 m, 1.0 m, 1.5 m, and 2.0 m were planned for the shooting to provide corresponding pixel conversion ratios at the experimental level.

4.2. Crack Quantification Principle. Based on the identified binary images of cracked components, it is important to calculate the commonly used crack information such as length and maximum width, et al. Since the number of pixels was used as the quantification value of cracks in the identified binary image, the following two steps were used to achieve crack information computation.

(1) Defining crack information computation method, as shown in equation (7).

$$Y_m = \phi \cdot X_{\mathrm{p}} \tag{7}$$

where Y_{m} and X_{p} represent the crack dimension in mm and pixel, respectively; ϕ is the conversion parameter (in mm/pixel).

(2) Calculating the conversion parameter ϕ using the targeted labeling method, as shown in equation (8).

$$\phi = D/d \tag{8}$$

where D denotes the dimension (in mm) of the marker on the components; d represents the pixel dimension of the marker occupied in the whole component image, in pixel.

The use of the targeted marking method meets the requirements of shooting from any angle and is no longer limited to frontal shooting. For ancient wooden components, it is more convenient. The width of beams, columns, and other wooden components can be used as reference markers. For object distances of 0.3 m, 0.5 m, 0.8 m, 1.0 m,

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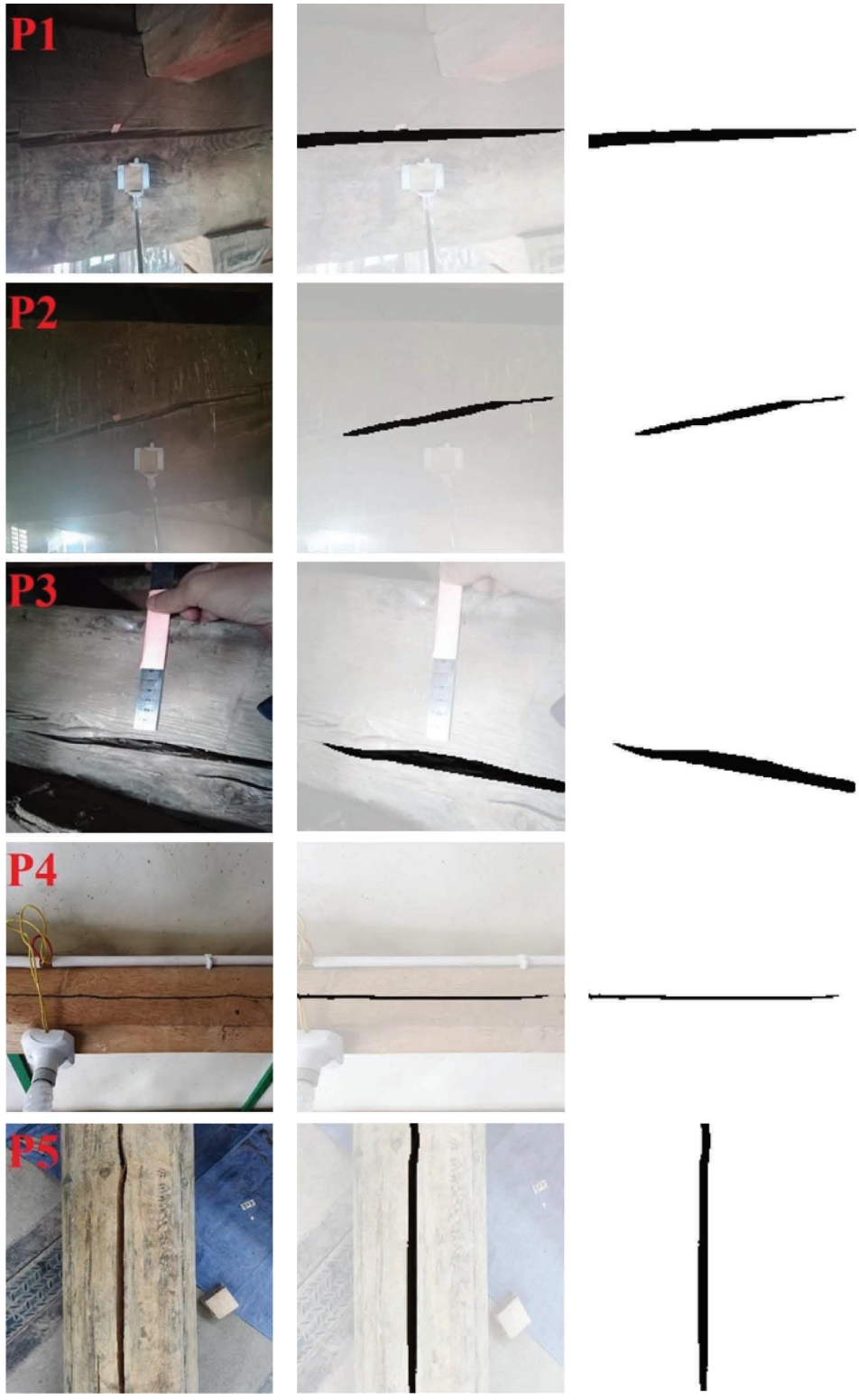


Figure 12: Test image crack recognition results. (a) Original image. (b) Identified crack image. (c) Identified binary image of crack.

1.5 m, and 2.0 m for the two-column specimens, the conversion ratios were averagely determined to be 0.025, 0.049, 0.085, 0.11, 0.174, and 0.236, respectively.

4.3. Crack Identification Results. Based on the captured images of wooden components, the ATM-Crack-U-net segmentation network proposed was first used and the

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FIGURE 13: Comparison of crack identification maps using multiple methods. (a) Original image. (b) WSA. (c) OSTU. (d) Canny. (e) ATM-crack U-net model.

optimal weight file ep095-loss0.167-val_loss0.169 was used for crack segmentation of the wooden component. The segmented images under different object distances are shown in Figures 15 and 16.

It can be seen that as the shooting distance increased, the crack recognition effect gradually deteriorated. When the object distance was less than 1 m, crack segmentation and recognition were better, and when it was greater than 1 m, crack segmentation and recognition were almost ineffective. It can be concluded that in the process of collecting images for identifying cracks in ancient wooden components, the shooting distance should be less than 1 m to ensure accurate identification and quantification of cracks.

4.4. Crack Quantification Method. After model segmentation and recognition for wood component crack images at different object distances, binary images containing only cracks were used for crack quantification calculation, including the length and maximum width of the cracks.

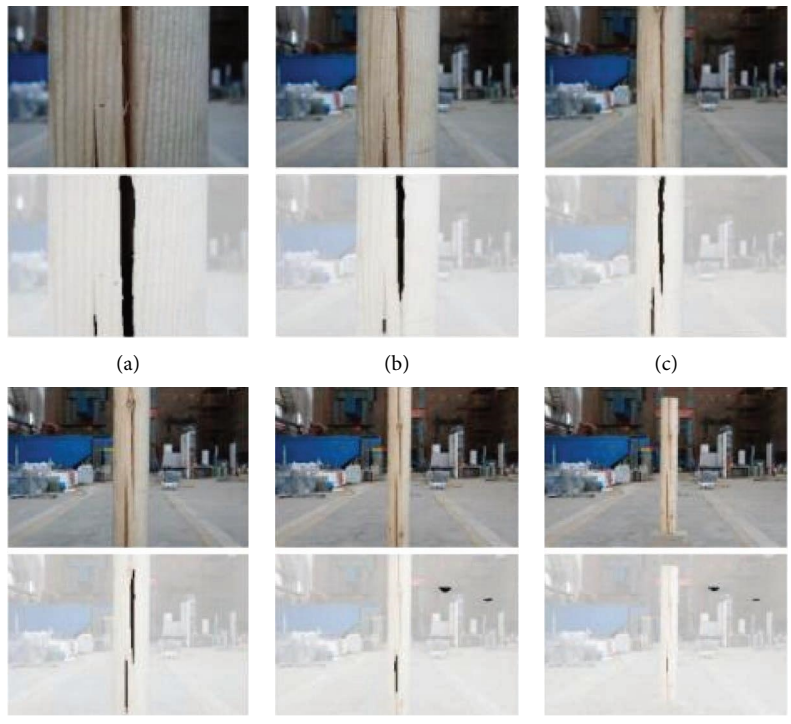
4.4.1. Crack Length. By indexing black pixels in binary images from one end to the other, the pixel value with the most cumulative number is the crack length pixel value L. Assuming that there are n black pixel points in the crack from one end to the other, and the coordinates of adjacent two-pixel points are (x_i, y_i) and (x_{i-1}, y_{i-1}), and the pixel distance between the two points is

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(a) (b) (c)

FIGURE 14: Sample image acquisition and manual measurement. (a) Cracked specimens. (b) Camera shooting. (c) Manual measurement.



(a) (b) (c)

(d) (e) (f)

FIGURE 15: Identification results of "01" specimen. (a) $u = 300$ mm. (b) $u = 500$ mm. (c) $u = 800$ mm. (d) $u = 1000$ mm. (e) $u = 1500$ mm. (f) $u = 2000$ mm.

$$L = \sum_{i=1}^{n} \sqrt{(x_i - x_{i-1})^2 + (y_i - y_{i-1})^2}. \quad (9)$$

The pixel values of crack length, manually measured actual distance values, and predicted values are calculated and shown in Table 3.

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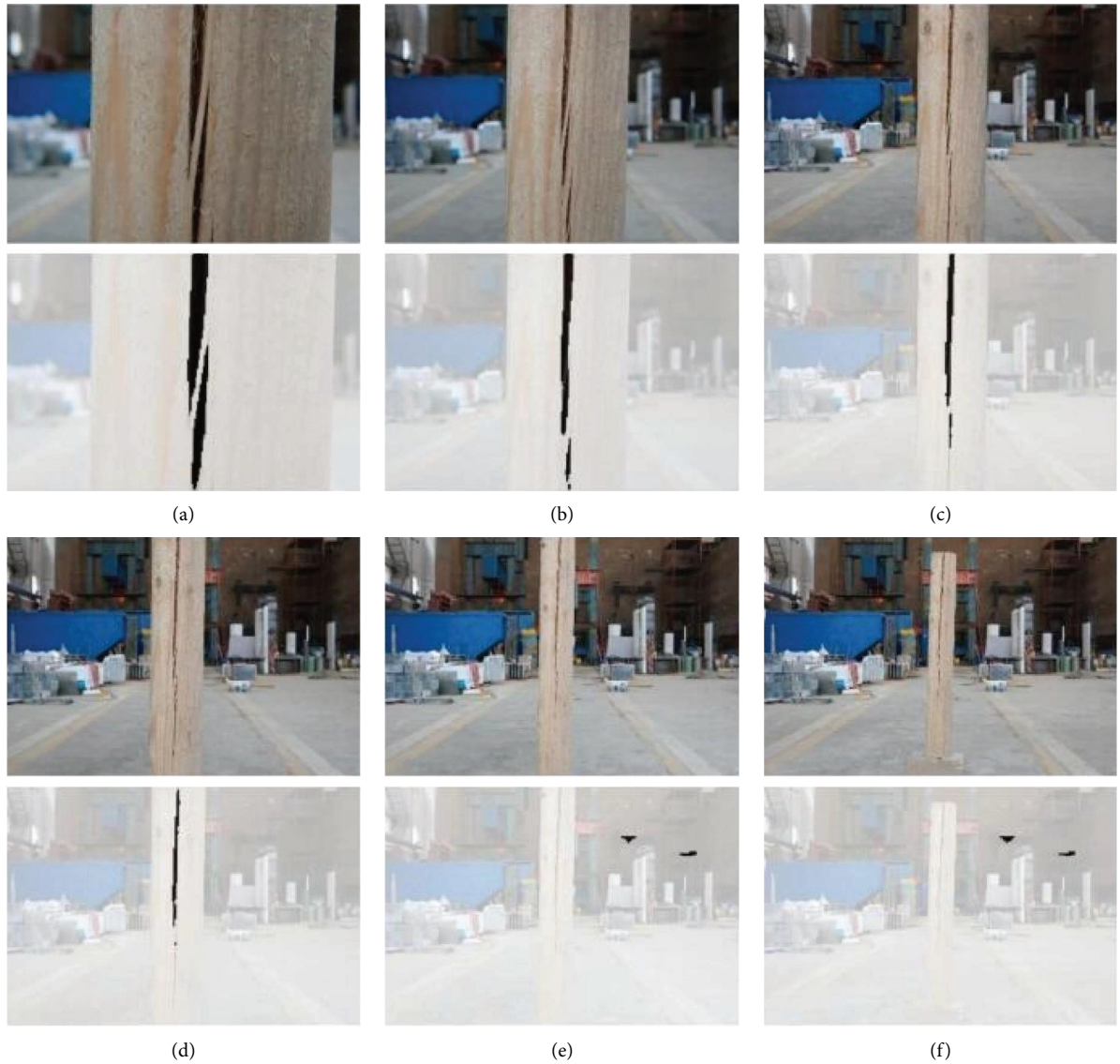


FIGURE 16: Identification results of "02" specimen. (a) $u = 300$ mm. (b) $u = 500$ mm. (c) $u = 800$ mm. (d) $u = 1000$ mm. (e) $u = 1500$ mm. (f) $u = 2000$ mm.

TABLE 3: Quantitative analysis of crack length.

Component no.	Pixel value (Pixe)	Actual distance (mm)	Predicted value (mm)	Error (%)
01–300	3920	82.6	97.8	18.42
01–500	3055	142.6	149.2	4.90
01–800	2937	233.7	248.4	6.28
01–1000	2286	236.2	252.3	6.82
02–300	3821	85.1	95.3	12.04
02–500	3771	177.8	184.1	3.56
02–800	3480	271.8	294.3	8.28
02–1000	2282	264.2	251.9	−4.66

It can be seen from Table 3 that when the shooting distance was 500 mm, the model had the smallest error between the predicted crack length and the actual manual measurement value. The recognition errors of the target labeling method for specimens "01" and "02" were 4.90% and 3.56%, respectively. When the shooting distance gradually increased from 500 mm to 1000 mm, the calculation errors increased as the object distance increases, indicating that the larger the shooting distance, the lower the accuracy. Comparing and analyzing the different shooting distances of

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Table 4: Quantitative analysis of maximum crack width.

Component no.	Pixel value (Pixe)	Actual value (mm)	Predicted value (mm)	Error (%)
01–300	306	7.5	7.6	1.33
01–500	188	9.0	9.2	2.22
01–800	141	11.2	11.9	6.25
01–1000	117	12.8	12.9	0.78
02–300	272	6.5	6.8	4.62
02–500	188	8.4	9.2	9.52
02–800	95	8.6	8.0	−6.98
02–1000	73	8.6	8.1	−5.81

two wooden components, the proposed crack length method was verified suitable for quantitative calculation of crack length in wooden components.

4.4.2. Crack Maximum Width. Cracks have two boundaries, the pixel sets of them can be defined as $A = [A_1\ A_2 \cdots A_i \cdots A_m]$ and $B = [B_1\ B_2 \cdots B_i \cdots B_n]$, respectively. The pixel coordinates of a point on boundaries A and B are (x_i, y_i) and (x_j, y_j). For a certain pixel point $A_i(x_i, y_i)$ on boundary A, the calculation of the corresponding crack width is to find the minimum value between the point A_i and any pixel point on boundary $\underline{B}$ (x_j, y_j), see equation (10).

$$W_i = \min\left(\sqrt{\left(x_i - x_j\right)^2 + \left(y_i - y_j\right)^2}\right) (i = 1, 2, \ldots, m;\ j == 1, 2, \ldots, n). \tag{10}$$

The maximum width for a certain crack can be calculated by finding the maximum value through equation (11).

$$W_{\max} = \max\left(W_i\right). \tag{11}$$

The pixel values of crack length, manually measured actual distance values, and predicted values are shown in Table 4. It can be seen that when the shooting distances of specimens "01" and "02" were within 1 m, the measured maximum local crack widths were 12.8 mm and 8.6 mm, respectively. When the shooting angle was 0.5 mm, the maximum width of cracks quantified by the proposed model for two specimens had an error of less than 10%. Through the comprehensive analysis of the quantification of crack length and maximum width, it can be concluded that the optimal shooting distance was 0.5 m (i.e., the object distance was equal to 0.5 m) with high quantification accuracy of crack length and width.

5. Conclusion

(1) The distribution patterns and morphological characteristics of cracks in wooden components were analyzed, and the cracked wooden components' dataset was established.

(2) A deep CNN model based on the VGG16-Unet network was established for intelligent identification of cracks in ancient wooden components. It concluded that the optimal initial learning rate takes the value of 10^{-4}, batch size of 6, and epoch of 100, achieving the average accuracy of 78.8% for the model in identifying wooden component cracks.

(3) Four segmentation algorithms were adopted to identify images with different crack characteristics. WSA and OTSU algorithms had poor recognition performance and were greatly affected by lighting and background. The Canny operator has a high level of recognition noise and blurry crack contours. The proposed model can segment smooth crack contours, effectively solving the problem of crack recognition in ancient wooden components.

(4) A method for quantifying cracks in wooden components based on image processing technology was proposed. The crack indicators were calculated using the target marking method, and the relationship between object distance and predicted results was analyzed. The results showed that the pixel conversion ratio values obtained by the target marking method were linearly positively correlated with object distance and can effectively identify cracks in wooden components within 1 m, with 0.5 m being the optimal object distance.

(5) The proposed crack recognition model can effectively identify apparent cracks, yet it still cannot provide crack depth information. Therefore, it is necessary to establish a correlation model between crack depth and apparent width and then construct a quantitative recognition method for three-dimensional crack morphology based on the deep learning technique.

Data Availability

The data used to support the findings of this study are included within the article.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

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Supplementary Materials

Supplemental figure 1: Loss values for different batch sizes when $Lr = 10^{-4}$. (a) Batch size = 2; (b) Batch size = 4; (c) Batch size = 6; (d) Batch size = 8. Supplemental figure 2: Loss values for different batch sizes when $Lr = 10^{-5}$. (a) Batch size = 2; (b) Batch size = 4; (c) Batch size = 6; (d) Batch size = 8. Supplemental figure 3: Loss values for different batch sizes when $Lr=10^{-6}$. (a) Batch size = 2; (b) Batch size = 4; (c) Batch size = 6; (d) Batch size = 8. Supplemental figure 4: Comparison of the optimal loss values between the training set and the validation set. (a) Training set; (b) Testing set. Supplemental figure 5: Predicted difference between different batch sizes when $Lr = 10^{-4}$. (a) Batch size = 2; (b) Batch size = 4; (c) Batch size = 6; (d) Batch size = 8. Supplemental figure 6: Predicted difference between different batch sizes when $Lr = 10^{-5}$. (a) Batch size = 2; (b) Batch size = 4; (c) Batch size = 6; (d) Batch size = 8. Supplemental figure 7: Predicted difference between different batch sizes when $Lr = 10^{-6}$. (a) Batch size = 2; (b) Batch size = 4; (c) Batch size = 6; (d) Batch size = 8. Supplemental figure 8: Comparison of the best F1-score between the training set and the validation set. (a) Training set; (b) Testing set. (*Supplementary Materials*)

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